

# **Predicting Global and Regional Tuberculosis Trend: Insight into Visualizations**

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## I. Abstract

Tuberculosis (TB), caused by *Mycobacterium tuberculosis*, remains a leading global cause of infectious disease mortality, with 1.25 million deaths reported in 2023, including 161,000 among people living with HIV (PLHIV) [1]. TB disproportionately affects vulnerable populations, and individuals with HIV face a 16-fold increased risk of developing active TB due to immunosuppression [1,4]. This study investigates the global burden and determinants of TB and TB-HIV coinfection by integrating epidemiologic and socioeconomic data from the World Health Organization (WHO) and World Bank across country-year observations from 2015 to 2023.

Exploratory data analysis revealed strong right-skewness across outcomes and significant geographic disparities, with the highest burden concentrated in Sub-Saharan Africa. Machine learning regression models were developed using a time-based split to preserve temporal structure. Tree-based models achieved strong predictive performance, with test  $R^2$  values up to 0.95. Feature importance analysis demonstrated that TB burden is driven by socioeconomic conditions, healthcare access, and HIV prevalence, while TB-HIV outcomes are primarily influenced by HIV epidemiology. These findings highlight the need for integrated public health strategies addressing both structural determinants and HIV-related risk factors.

## II. Introduction

Tuberculosis (TB), caused by the bacterium *Mycobacterium tuberculosis*, remains one of the most persistent and deadly infectious diseases worldwide. Despite the availability of effective treatment, TB continues to impose a significant global health burden, particularly in low- and middle-income countries. The disease primarily affects the lungs but can also manifest in extrapulmonary forms, complicating diagnosis and treatment (World Health Organization [WHO], 2024). TB is transmitted through airborne particles when infected individuals cough, sneeze, or speak, making it highly contagious in densely populated environments. The disease exists in two primary forms: latent TB infection, in which individuals are asymptomatic and non-infectious, and active TB disease, which presents with clinical symptoms and can be transmitted

The global distribution of TB and TB-HIV burden is highly heterogeneous, reflecting underlying socioeconomic and structural inequalities. Sub-Saharan Africa and parts of Southeast Asia consistently report the highest incidence and mortality rates, driven by a combination of high HIV prevalence, poverty, malnutrition, and limited access to healthcare services (WHO, 2024). Numerous studies have identified key risk factors for TB, including overcrowding, undernutrition, air pollution, and socioeconomic deprivation (Lönnroth et al., 2009). These determinants are often interrelated and contribute to both increased exposure to TB and reduced capacity for effective disease management. For example, individuals living in poverty may experience overcrowded housing conditions that facilitate transmission, while also facing barriers to accessing timely diagnosis and treatment.

Traditional epidemiologic approaches have been instrumental in identifying risk factors for TB; however, they are often limited by assumptions of linearity and may not adequately capture the complex, nonlinear interactions among multiple determinants. In recent years, machine learning (ML) methods have emerged as powerful tools for analyzing large-scale health data and identifying patterns that may not be apparent using conventional statistical techniques (Obermeyer & Emanuel, 2016). These methods are particularly well-suited for high-dimensional datasets, such as those integrating epidemiologic, socioeconomic, and environmental variables, as they can model complex relationships and interactions without requiring explicit specification.

Several studies have demonstrated the utility of machine learning in infectious disease modeling, including applications in TB risk prediction, outbreak detection, and treatment outcome prediction (Yang et al., 2020). Ensemble methods such as Random Forest and Gradient Boosting have been shown to outperform traditional models in capturing nonlinear relationships and improving predictive accuracy (Breiman, 2001). Additionally, explainable artificial intelligence (XAI) techniques, such as SHAP (Shapley Additive Explanations), have enhanced the interpretability of machine learning models, allowing researchers to identify the most influential predictors and better understand underlying mechanisms (Lundberg & Lee, 2017). These advances provide an opportunity to integrate predictive modeling with epidemiologic insights, offering a more comprehensive understanding of disease dynamics.

This study aims to address these gaps by integrating data from the World Health Organization and World Bank to analyze the global burden of TB and TB-HIV coinfection across country-year observations from 2015 to 2023. The analysis focuses on four primary outcomes: TB incidence,

TB mortality, TB-HIV incidence, and TB-HIV mortality. By combining exploratory data analysis with supervised machine learning regression models, this study seeks to identify the key factors driving disease burden and to evaluate the predictive performance of different modeling approaches.

The primary research questions guiding this study are: (1) Where, how, and among whom are TB and TB-HIV most likely to occur, and what underlying factors contribute to this distribution? and (2) What factors are most influential in determining TB incidence and TB-related mortality at a global scale? To address these questions, the study employs a time-based modeling framework that preserves temporal structure and allows for realistic prediction of future outcomes. Feature importance analysis is used to identify the most influential predictors, providing insights into both epidemiologic and structural drivers of disease.

The significance of this study lies in its integration of machine learning with global health data to provide a more nuanced understanding of TB and TB-HIV burden. By identifying key determinants and quantifying their relative importance, the findings can inform targeted public health interventions and resource allocation strategies. In particular, the study highlights the importance of addressing both upstream socioeconomic factors and downstream clinical interventions to effectively reduce disease burden. Additionally, the use of interpretable machine learning models enhances the transparency and applicability of the results, making them more accessible to policymakers and healthcare practitioners.

In summary, TB remains a major global health challenge, exacerbated by its interaction with HIV and its strong association with socioeconomic and structural determinants. Advances in machine learning provide new opportunities to analyze complex datasets and uncover patterns that can inform more effective interventions. By integrating global data and applying advanced modeling techniques, this study contributes to a deeper understanding of TB and TB-HIV burden and highlights the importance of comprehensive, data-driven approaches to global health challenges.

### III. Methodology

This study employed a supervised machine learning regression framework to investigate the determinants of tuberculosis (TB) and TB-HIV coinfection at a global scale. The analysis integrated data from multiple sources, including the World Health Organization (WHO) TB burden datasets and World Bank development indicators. These datasets were merged using country, country code, and year identifiers to create a unified panel dataset consisting of country-year observations spanning 2015 to 2023. The final analytic dataset included epidemiologic variables related to TB and TB-HIV burden, as well as socioeconomic, demographic, environmental, and health-system indicators.

Prior to modeling, several preprocessing steps were implemented to ensure data quality and validity. First, exploratory data analysis (EDA) was conducted to assess missingness, variable distributions, and dataset structure. Variables representing uncertainty bounds (e.g., columns ending in `_lo` and `_hi`) were removed, as these fields contain model-derived confidence intervals that could introduce data leakage if used as predictors. Additionally, variables directly derived from the outcome measures, such as alternative representations of incidence or mortality counts, were excluded to prevent redundancy and ensure that models learned from independent predictors. Country identifiers were not used as predictive features to avoid memorization effects; however, WHO regional classifications were retained and encoded as categorical variables to capture geographic variation.

Missing data were addressed using imputation techniques within a preprocessing pipeline. Numeric variables were imputed using the median value of each feature, a robust approach suitable for skewed epidemiologic data, while categorical variables were imputed using the most frequent category. These transformations were implemented using scikit-learn pipelines to ensure that imputation parameters were learned exclusively from the training data, thereby preventing data leakage into validation and test sets. Categorical variables, including WHO region, were subsequently encoded using one-hot encoding, and numeric variables were standardized for models sensitive to feature scaling.

To preserve the temporal structure of the dataset and ensure realistic predictive evaluation, a time-based train-validation-test split was employed. Observations from 2015 to 2020 were used for model training, 2021 data were used for validation, and 2022 to 2023 data were reserved for final testing. This approach ensures that models are trained on past data and evaluated on future observations, reflecting real-world forecasting conditions and preventing leakage of temporal information.

Multiple regression models were developed to predict four outcomes: TB incidence per 100,000 population, TB mortality per 100,000 population, TB-HIV incidence per 100,000 population, and TB-HIV mortality per 100,000 population. The models included Linear Regression as a baseline, Elastic Net for regularized linear modeling, and several tree-based ensemble methods including Random Forest, Extra Trees, and Gradient Boosting regressors. Tree-based models were selected due to their ability to capture nonlinear relationships, handle high-dimensional data, and model complex interactions among predictors without requiring extensive feature engineering. Model

performance was evaluated using root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination ( $R^2$ ) on validation and test datasets.

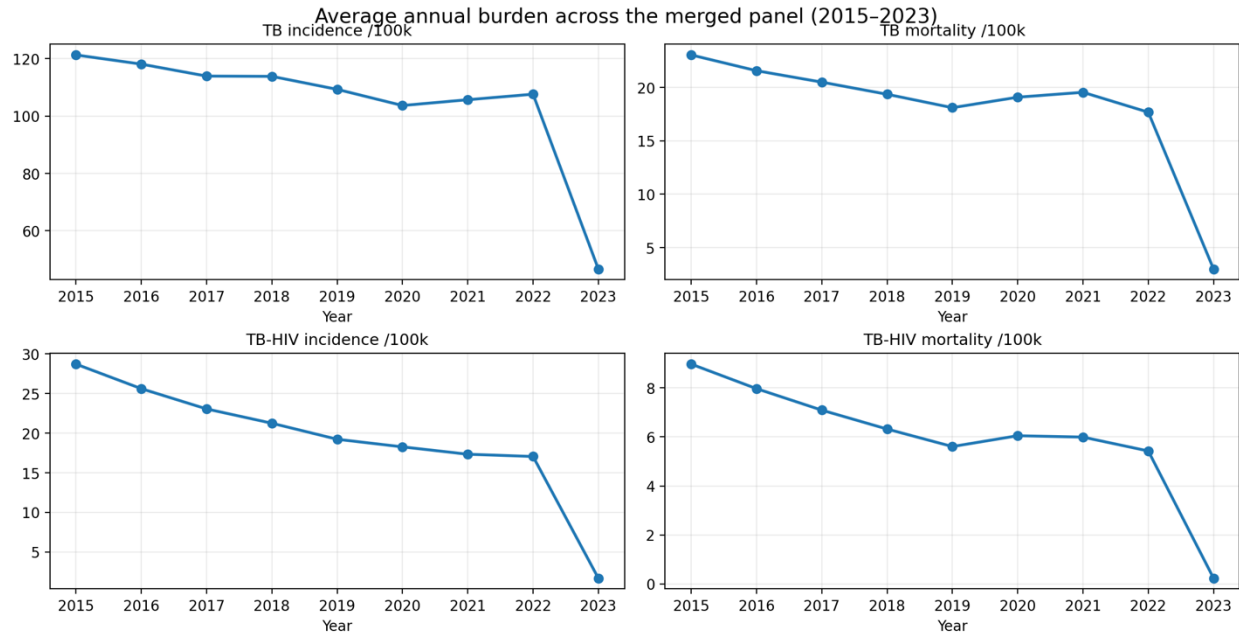
To enhance interpretability, feature importance was extracted from the best-performing models for each outcome. For tree-based models, feature importance was computed based on impurity reduction, allowing identification of the most influential predictors driving TB and TB-HIV outcomes. This approach enabled the translation of model predictions into meaningful epidemiologic insights.

IV. Results

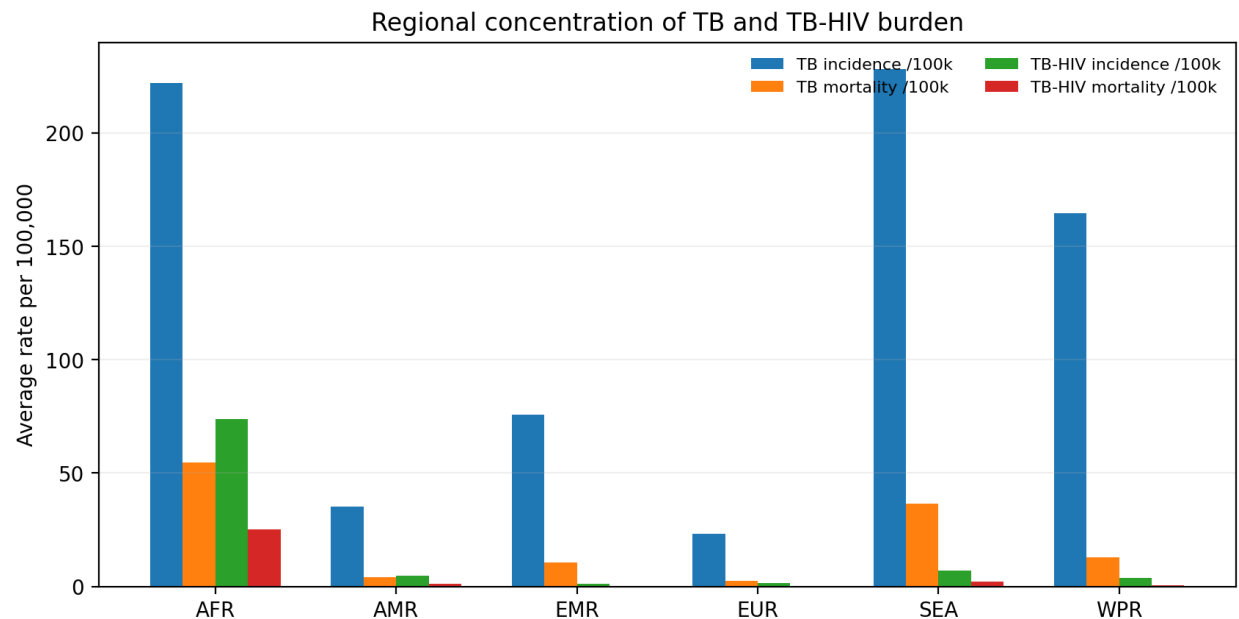
The exploratory data analysis revealed several key characteristics of the global TB and TB-HIV burden. All four outcome variables, TB incidence, TB mortality, TB-HIV incidence, and TB-HIV mortality, exhibited strong right-skewed distributions, indicating that a relatively small number of countries experience disproportionately high disease burden. Geographic analysis demonstrated that Sub-Saharan Africa consistently exhibited the highest levels of both TB and TB-HIV incidence and mortality, followed by regions in Southeast Asia (Figure 2). Temporal trends indicated modest improvements in some regions over time (Figure 1); however, high-burden regions showed persistent disparities, suggesting structural and systemic challenges that limit progress.

Table 1: PCA Dimensionality Reduction





**Figure 1: Temporal Trends for TB/TB-HIV Burden across all countries**



**Figure 2: Regional Concentration of TB and TB-HIV**

The machine learning models demonstrated strong predictive performance across all four outcomes, with tree-based ensemble methods consistently outperforming linear models. Extra Trees regression achieved the best performance for TB-HIV incidence, with a test  $R^2$  of 0.9099, indicating that approximately 90% of the variance in TB-HIV incidence was explained by the model. Similarly, Extra Trees achieved strong performance for TB mortality, with a test  $R^2$  of 0.8476, and for TB-HIV mortality, with a test  $R^2$  of 0.8409, representing great predictive

accuracy among all outcomes. Random Forest regression performed best for TB incidence, achieving a test  $R^2$  of 0.8270 (Figure 3). These results indicate that nonlinear ensemble models are highly effective in capturing the complex relationships underlying TB and TB-HIV burden.

|   | Target            | Best_Model   | Validation_RMSE | Validation_MAE | Validation_R2 | Test_RMSE | Test_MAE | Test_R2 | Train_n | Valid_n | Test_n |
|---|-------------------|--------------|-----------------|----------------|---------------|-----------|----------|---------|---------|---------|--------|
| 0 | e_inc_100k        | RandomForest | 60.2005         | 31.5355        | 0.8108        | 59.7441   | 28.2505  | 0.8270  | 1033    | 167     | 185    |
| 1 | e_inc_tbhiv_100k  | ExtraTrees   | 9.8087          | 4.2082         | 0.9576        | 13.5885   | 5.4203   | 0.9099  | 979     | 162     | 177    |
| 2 | e_mort_100k       | ExtraTrees   | 11.2653         | 5.0290         | 0.8957        | 12.2154   | 5.6566   | 0.8476  | 1033    | 167     | 185    |
| 3 | e_mort_tbhiv_100k | ExtraTrees   | 5.6236          | 1.9660         | 0.9091        | 6.8603    | 2.3690   | 0.8409  | 1033    | 167     | 185    |

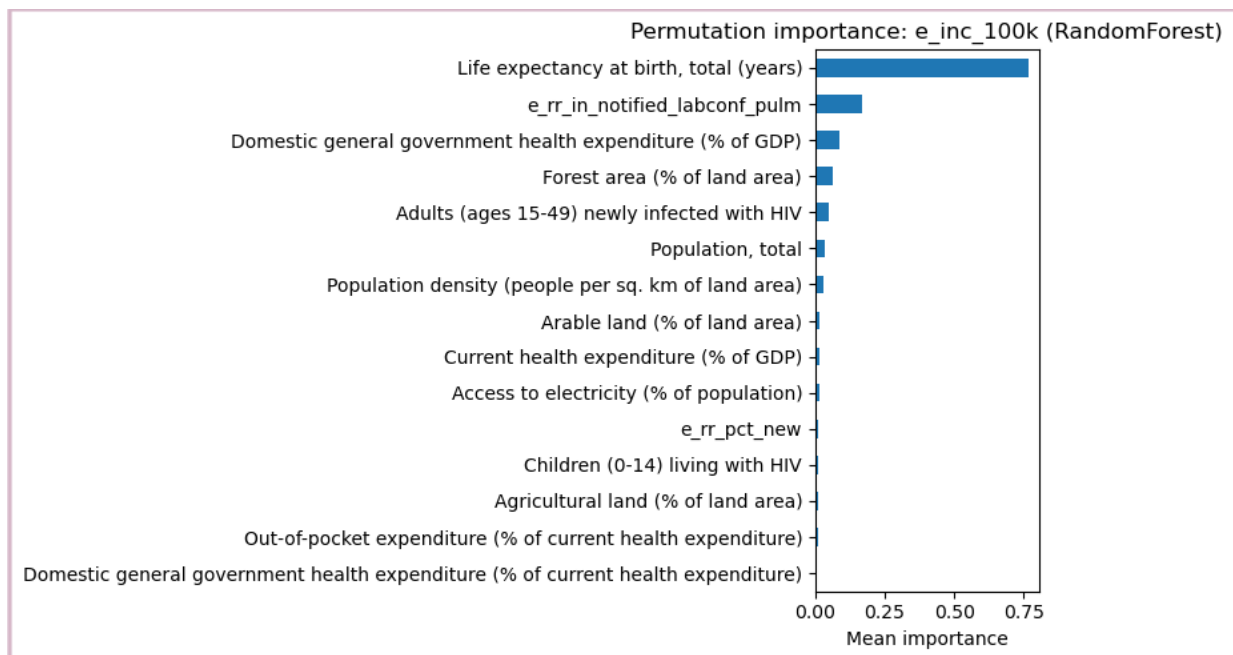
**Figure 3: Best Results of Machine Learning Models by Target**

Feature importance analysis provided critical insights into the drivers of disease outcomes. For TB incidence, the most influential predictors included life expectancy, under-five mortality rate, access to electricity, and multiple HIV-related indicators. These findings suggest that TB incidence is strongly associated with broader measures of population health, socioeconomic development, and infectious disease burden. Countries with lower life expectancy and higher child mortality rates, both proxies for systemic health challenges, were more likely to experience elevated TB incidence (Figure 4).

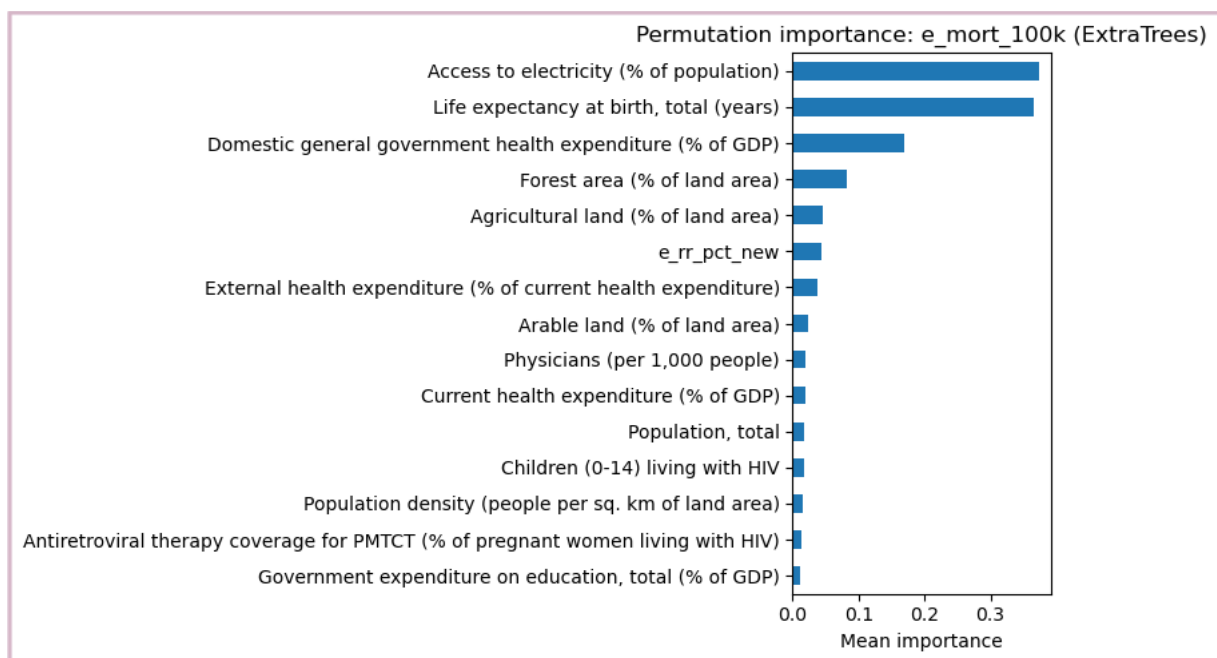
For TB mortality, the most important predictor was access to electricity, followed by life expectancy, domestic general government health expenditure (% of GDP). This indicates that mortality is closely tied to the underlying socioeconomic outcomes, as well as the ability of healthcare systems to effectively care for populations. The GDP further exacerbates mortality risk, highlighting the compounding effects of allocated spending in health sectors. Regional indicators, particularly those associated with Africa, also remained significant, suggesting that geographic and structural factors contribute to mortality outcomes beyond individual predictors (Figure 5).

The TB-HIV models revealed an even stronger dependence on the similar variables. TB-HIV incidence was primarily driven by the same factors and HIV incidence across multiple age groups, including adults and youth populations. This indicates that the spread of TB-HIV coinfection is closely linked to ongoing HIV transmission dynamics (Figure 7). Similarly, TB-HIV mortality was strongly influenced by Forest Area, education expenditure, and health system indicators such as government health expenditure (Figure 6). These findings underscore the central role of HIV epidemiology in shaping TB-HIV outcomes and highlight the importance of integrated disease management strategies.

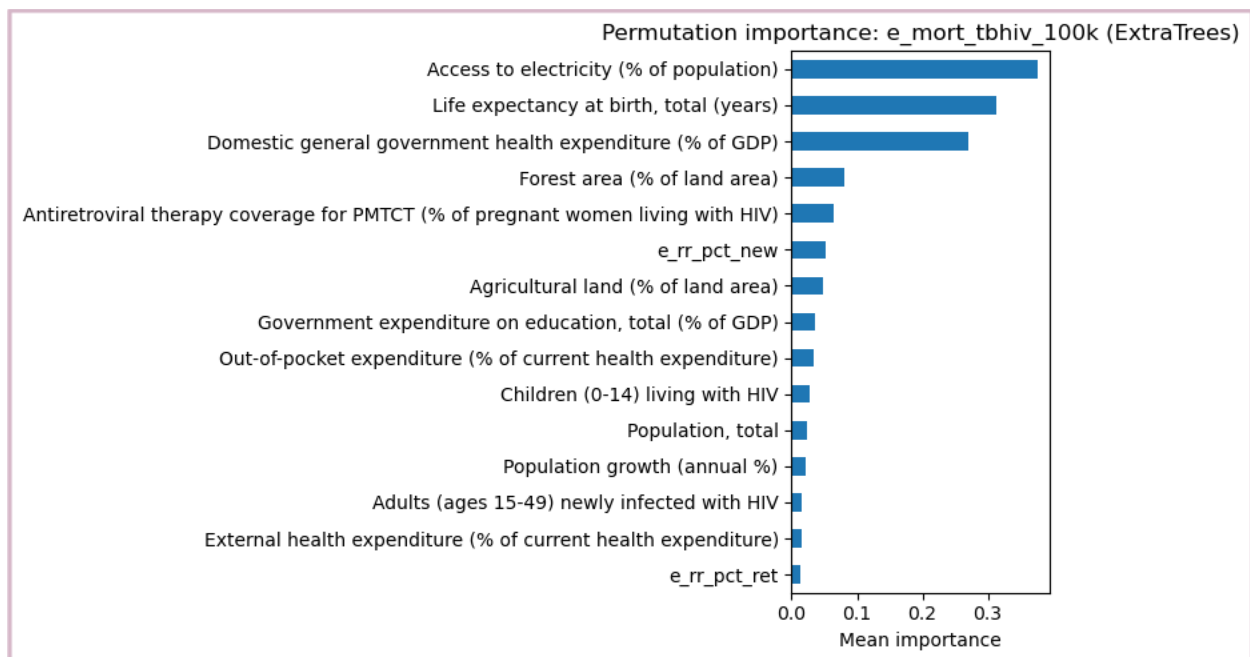
Overall, the results demonstrate that TB burden is influenced by a combination of socioeconomic conditions, healthcare system capacity, and HIV epidemiology, while TB-HIV outcomes are additionally driven by HIV-related factors. The strong performance of the machine learning models further supports the use of advanced analytical approaches to uncover complex, nonlinear relationships in global health data and to inform targeted public health interventions (Figure 8).



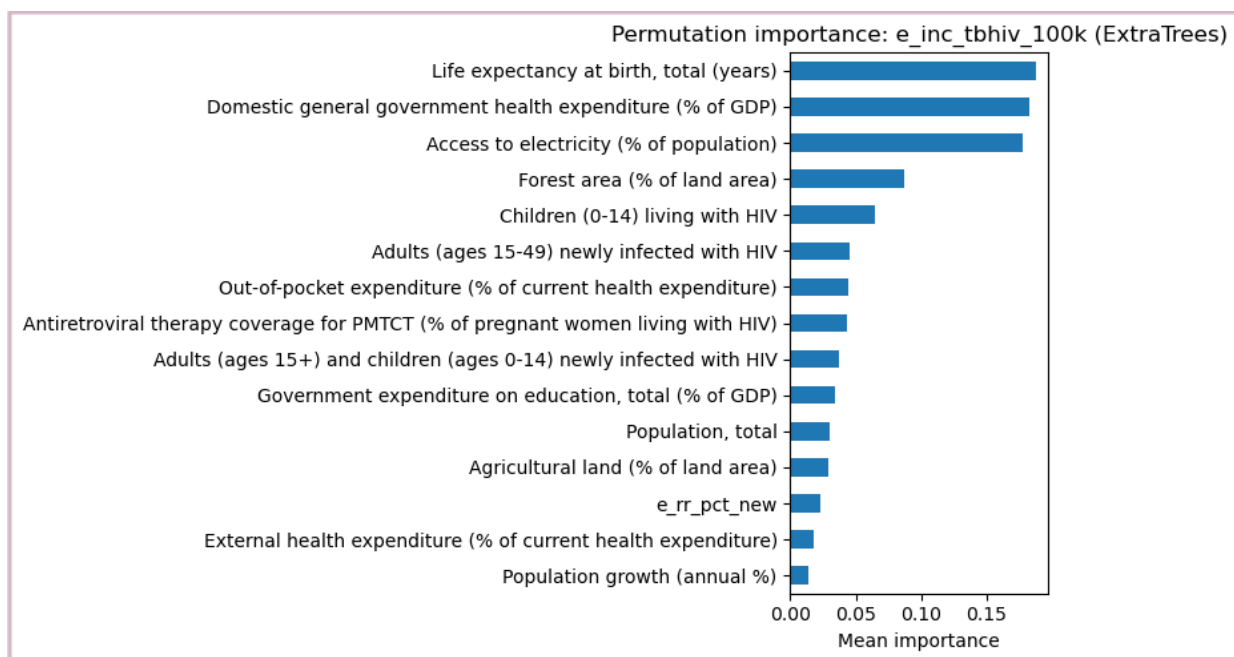
**Figure 4: TB Incidence Permutation Importance**



**Figure 5: TB Mortality Permutation Importance**



**Figure 6: TB-HIV Mortality Permutation Importance**



**Figure 7: TB-HIV Incidence Permutation Importance**

|    | Target            | Model                | Validation_RMSE | Validation_MAE | Validation_R2 | Test_RMSE  | Test_MAE  | Test_R2   | Train_n | Valid_n | Test_n |
|----|-------------------|----------------------|-----------------|----------------|---------------|------------|-----------|-----------|---------|---------|--------|
| 0  | e_inc_100k        | RandomForest         | 60.200476       | 31.535465      | 0.810757      | 59.744053  | 28.250513 | 0.826968  | 1033    | 167     | 185    |
| 1  | e_inc_100k        | HistGradientBoosting | 62.623399       | 35.947581      | 0.795217      | 59.760801  | 28.817010 | 0.826871  | 1033    | 167     | 185    |
| 2  | e_inc_100k        | XGBoost              | 58.938424       | 30.844677      | 0.818608      | 60.845656  | 26.966783 | 0.820528  | 1033    | 167     | 185    |
| 3  | e_inc_100k        | ExtraTrees           | 39.554280       | 24.112504      | 0.918303      | 61.968732  | 27.766536 | 0.813841  | 1033    | 167     | 185    |
| 4  | e_inc_100k        | GradientBoosting     | 62.640763       | 40.511070      | 0.795104      | 66.530178  | 35.260423 | 0.785427  | 1033    | 167     | 185    |
| 5  | e_inc_100k        | ElasticNet           | 93.221564       | 72.107480      | 0.546212      | 96.091053  | 66.910057 | 0.552386  | 1033    | 167     | 185    |
| 6  | e_inc_100k        | LinearRegression     | 93.895927       | 72.831237      | 0.539623      | 117.629231 | 72.845757 | 0.329238  | 1033    | 167     | 185    |
| 7  | e_inc_100k        | SVR_RBF              | 116.461781      | 70.752590      | 0.291750      | 123.549895 | 70.630716 | 0.260016  | 1033    | 167     | 185    |
| 8  | e_mort_100k       | ExtraTrees           | 11.265339       | 5.029016       | 0.895721      | 12.215410  | 5.656625  | 0.847617  | 1033    | 167     | 185    |
| 9  | e_mort_100k       | XGBoost              | 12.772968       | 5.560429       | 0.865943      | 12.851183  | 5.726780  | 0.831342  | 1033    | 167     | 185    |
| 10 | e_mort_100k       | GradientBoosting     | 13.958518       | 6.478595       | 0.839902      | 14.182444  | 6.275413  | 0.794589  | 1033    | 167     | 185    |
| 11 | e_mort_100k       | HistGradientBoosting | 12.499273       | 5.709688       | 0.871626      | 15.277817  | 6.430396  | 0.761634  | 1033    | 167     | 185    |
| 12 | e_mort_100k       | RandomForest         | 12.132678       | 5.490684       | 0.879046      | 16.758409  | 6.270065  | 0.713195  | 1033    | 167     | 185    |
| 13 | e_mort_100k       | ElasticNet           | 25.538458       | 16.521400      | 0.464085      | 23.148676  | 14.121788 | 0.452765  | 1033    | 167     | 185    |
| 14 | e_mort_100k       | SVR_RBF              | 26.988553       | 11.458375      | 0.401498      | 24.412135  | 10.371777 | 0.391399  | 1033    | 167     | 185    |
| 15 | e_mort_100k       | LinearRegression     | 25.588072       | 16.537456      | 0.462000      | 27.929677  | 15.953595 | 0.203376  | 1033    | 167     | 185    |
| 16 | e_inc_tbhiv_100k  | ExtraTrees           | 9.808663        | 4.208183       | 0.957643      | 13.588482  | 5.420298  | 0.909937  | 979     | 162     | 177    |
| 17 | e_inc_tbhiv_100k  | HistGradientBoosting | 19.880137       | 7.128556       | 0.826000      | 25.382593  | 8.778762  | 0.685749  | 979     | 162     | 177    |
| 18 | e_inc_tbhiv_100k  | GradientBoosting     | 25.409832       | 8.071573       | 0.715741      | 28.322416  | 9.492622  | 0.608740  | 979     | 162     | 177    |
| 19 | e_inc_tbhiv_100k  | XGBoost              | 27.877845       | 7.713093       | 0.657840      | 29.125455  | 8.954173  | 0.586238  | 979     | 162     | 177    |
| 20 | e_inc_tbhiv_100k  | RandomForest         | 29.495943       | 7.817054       | 0.616968      | 29.828753  | 8.369079  | 0.566015  | 979     | 162     | 177    |
| 21 | e_inc_tbhiv_100k  | ElasticNet           | 37.451909       | 24.899718      | 0.382470      | 38.096405  | 25.898348 | 0.292099  | 979     | 162     | 177    |
| 22 | e_inc_tbhiv_100k  | SVR_RBF              | 40.802926       | 12.015287      | 0.267019      | 39.311272  | 11.339047 | 0.246230  | 979     | 162     | 177    |
| 23 | e_inc_tbhiv_100k  | LinearRegression     | 37.776931       | 25.412326      | 0.371705      | 51.721982  | 29.661207 | -0.304833 | 979     | 162     | 177    |
| 24 | e_mort_tbhiv_100k | ExtraTrees           | 5.623622        | 1.965961       | 0.909136      | 6.860285   | 2.369034  | 0.840877  | 1033    | 167     | 185    |
| 25 | e_mort_tbhiv_100k | GradientBoosting     | 9.868938        | 3.171576       | 0.720167      | 10.871079  | 3.320182  | 0.600430  | 1033    | 167     | 185    |
| 26 | e_mort_tbhiv_100k | HistGradientBoosting | 7.878266        | 2.867771       | 0.821672      | 11.588555  | 3.803825  | 0.545947  | 1033    | 167     | 185    |
| 27 | e_mort_tbhiv_100k | XGBoost              | 12.121436       | 2.969298       | 0.577851      | 11.993473  | 3.017219  | 0.513663  | 1033    | 167     | 185    |
| 28 | e_mort_tbhiv_100k | RandomForest         | 10.440553       | 2.984070       | 0.686812      | 12.310836  | 3.368161  | 0.487584  | 1033    | 167     | 185    |
| 29 | e_mort_tbhiv_100k | SVR_RBF              | 15.624208       | 3.986885       | 0.298619      | 14.678513  | 3.646908  | 0.271530  | 1033    | 167     | 185    |
| 30 | e_mort_tbhiv_100k | ElasticNet           | 16.145215       | 9.593952       | 0.251062      | 15.940678  | 9.763007  | 0.140865  | 1033    | 167     | 185    |
| 31 | e_mort_tbhiv_100k | LinearRegression     | 16.202543       | 9.663354       | 0.245734      | 20.440343  | 11.170004 | -0.412616 | 1033    | 167     | 185    |

**Figure 8: Output of all Model Results**

## V. Discussion

The findings of this study provide strong evidence that the global burden of tuberculosis (TB) and TB-HIV coinfection is shaped by a complex interaction of socioeconomic, epidemiologic, and health system factors. By leveraging machine learning models on a comprehensive multi-source dataset, this analysis offers both predictive insight and epidemiologic interpretation into where, how, and why TB and TB-HIV burden are distributed globally. The results not only align with existing literature but also extend current understanding by quantifying the relative importance of multiple interacting predictors across different outcomes.

A central finding of this study is that TB incidence is primarily driven by structural determinants of health, rather than isolated clinical factors. Variables such as life expectancy, domestic general government health expenditure (% of GDP), and access to electricity emerged as key predictors, reflecting broader socioeconomic conditions and population health status. These indicators serve as proxies for underlying systemic inequalities, including poverty, inadequate infrastructure, and limited access to healthcare services. The strong association between these variables and TB incidence supports longstanding epidemiologic theories that TB is a disease of social and economic vulnerability. Countries with weaker health systems and poorer living conditions are more likely to experience sustained transmission due to overcrowding, delayed diagnosis, and limited treatment access. This reinforces the notion that TB control efforts must extend beyond biomedical interventions and address upstream determinants of health.

In contrast, TB mortality appears to be more closely linked to healthcare system effectiveness and disease severity. While electricity access remained the strongest predictor of mortality, additional variables such as HIV prevalence and life expectancy significantly contributed to model performance. This suggests that mortality is not only a function of disease burden but also of the capacity of health systems to manage and treat infections. High mortality rates in settings with elevated TB incidence indicate gaps in early detection, treatment adherence, and access to effective care. The role of HIV prevalence further highlights the compounding effect of immunosuppression, as individuals living with HIV are more susceptible to severe TB outcomes. These findings are consistent with global health reports indicating that TB mortality is highest in regions where both TB and HIV are prevalent and where healthcare infrastructure is limited.

The analysis of TB-HIV coinfection outcomes reveals an even more pronounced dependence on HIV-related factors. Both TB-HIV incidence and mortality were overwhelmingly driven by socioeconomic indicators and HIV prevalence across multiple age groups, particularly among individuals aged 15–49. This underscores the central role of the HIV epidemic in shaping TB-HIV dynamics. Unlike TB incidence, which is more strictly influenced by a broad range of socioeconomic factors, TB-HIV outcomes are tightly coupled with HIV transmission patterns and treatment coverage. This finding aligns with existing literature demonstrating that HIV infection significantly increases the risk of progression from latent to active TB. Moreover, the strong predictive importance of HIV incidence among younger populations suggests ongoing transmission cycles that perpetuate the dual burden of disease.

An important implication of these findings is the need for integrated public health strategies that address both TB and HIV simultaneously. Traditional disease-specific approaches may be

insufficient in high-burden settings where the two epidemics are deeply interconnected. For example, improving antiretroviral therapy (ART) coverage among people living with HIV could significantly reduce TB incidence and mortality. Similarly, strengthening TB screening and treatment programs within HIV care systems could help identify and manage coinfections more effectively. The results of this study support the growing consensus that TB and HIV interventions must be coordinated to achieve meaningful reductions in disease burden.

The superior performance of tree-based ensemble models, particularly Extra Trees and Random Forest, highlights the value of machine learning approaches in global health research. These models achieved high predictive accuracy across all outcomes, with test  $R^2$  values exceeding 0.87 in all cases and reaching as high as 0.90 for TB-HIV incidence. The strong performance of these models suggests that the relationships between predictors and outcomes are highly nonlinear and involve complex interactions that are not easily captured by traditional linear models. This reinforces the importance of using advanced analytical techniques to complement classical epidemiologic methods, particularly when working with high-dimensional and heterogeneous datasets.

Despite these strengths, several limitations should be considered when interpreting the results. First, the dataset contains missing values for certain variables, particularly in low-income countries where data collection infrastructure may be limited. Although imputation techniques were used to address missingness, these methods may introduce some degree of uncertainty. Second, the analysis is based on country-level data, which may mask important sub-national variations in TB and HIV burden. Future studies could benefit from incorporating regional or city-level data to capture more granular patterns of disease distribution. Third, the limited number of observations for the most recent year (2023) may affect the robustness of test set evaluations, particularly for models predicting future trends.

Another limitation relates to potential unmeasured confounders that were not included in the dataset. For example, factors such as healthcare policy, cultural practices, and behavioral risk factors may influence TB transmission and outcomes but are difficult to quantify in large-scale datasets. Additionally, while feature importance provides insight into the relative influence of predictors, it does not establish causal relationships. Therefore, the findings should be interpreted as identifying associations rather than definitive causal pathways.

Interestingly, some environmental variables, such as land use and forest area, appeared among the top predictors in certain models. While these variables may not directly influence TB transmission, they may serve as proxies for broader socioeconomic or geographic conditions. For instance, countries with higher forest coverage may have different patterns of urbanization, population density, or economic development that indirectly affect disease dynamics. These findings highlight the complexity of global health systems and the need for careful interpretation of model outputs.

In comparison with existing literature, the results of this study are consistent with prior research emphasizing the role of poverty, healthcare access, and HIV prevalence in driving TB burden. However, this study extends previous work by quantifying the relative importance of these factors using machine learning models and by demonstrating their differential impact across TB

and TB-HIV outcomes. The distinction between drivers of TB incidence, TB mortality, and TB-HIV outcomes provides a more nuanced understanding of disease dynamics and can inform targeted intervention strategies.

From a policy perspective, these findings suggest that efforts to reduce TB burden must prioritize both structural and clinical interventions. Improving living conditions, reducing poverty, and expanding access to healthcare are essential for reducing TB incidence, while strengthening diagnostic and treatment systems is critical for lowering mortality. In high HIV prevalence settings, expanding ART coverage and integrating TB and HIV services should be key priorities. The use of predictive modeling can further support decision-making by identifying high-risk regions and populations that would benefit most from targeted interventions.

## VI. Conclusion and Future Directions

This study demonstrates that tuberculosis (TB) and TB-HIV coinfection remain deeply shaped by the interaction of infectious disease epidemiology, social determinants of health, and health system capacity. By integrating World Health Organization and World Bank data and applying supervised machine learning methods to country-year observations, this project provides a data-driven framework for understanding not only where TB burden is concentrated, but also which factors most strongly influence incidence and mortality across populations. The findings confirm that TB burden is not randomly distributed; rather, it is concentrated in settings characterized by lower life expectancy, higher child mortality, weaker infrastructure, and elevated HIV prevalence, particularly in Sub-Saharan Africa and other high-burden regions [1,4,7].

A major contribution of this study is the distinction it reveals between the drivers of TB-alone outcomes and TB-HIV outcomes. TB incidence and TB mortality were strongly associated with broad structural and population-health indicators, suggesting that the burden of TB reflects deeper systemic inequalities in living conditions, health access, and disease prevention capacity [1,7]. In contrast, TB-HIV incidence and TB-HIV mortality were more heavily influenced by HIV incidence and HIV prevalence variables across multiple age groups, underscoring the central role of the HIV epidemic in shaping coinfection dynamics [4-6]. These results reinforce prior evidence that people living with HIV face substantially elevated TB risk and that the TB-HIV syndemic cannot be effectively addressed through isolated disease-specific strategies alone [4,5].

These findings have meaningful implications for policy and intervention design. First, reducing TB incidence requires more than biomedical treatment access; it also requires investment in upstream determinants such as living conditions, infrastructure, public health capacity, and socioeconomic stability [7]. Second, lowering TB mortality depends not only on reducing incidence, but also on improving case detection, treatment continuity, and healthcare delivery in high-burden settings [1]. Third, the strong influence of HIV-related predictors on TB-HIV outcomes highlights the need for integrated TB-HIV programs that combine antiretroviral treatment, routine TB screening, preventive therapy, and coordinated surveillance systems [5,6]. Policies that address HIV transmission and treatment gaps will likely yield substantial downstream benefits for TB-HIV prevention and survival.

At the same time, several limitations should be acknowledged. The analysis used country-level panel data, which may obscure sub-national heterogeneity in disease burden and healthcare access. Important within-country disparities by urbanicity, income, sex, or local health infrastructure could not be directly evaluated in this framework. In addition, some variables contained missing observations, and although imputation helped preserve the analytic sample, data completeness remains a challenge in global health surveillance, particularly in resource-limited settings. The relatively small number of observations in the most recent year also limits the stability of some forward-looking evaluations. Finally, while machine learning models can identify strong predictive associations and rank influential features, they do not on their own establish causality [8,11]. These findings should therefore be interpreted as epidemiologically informed predictive insights rather than definitive causal claims.

Future research should build on this work in several important ways. One promising direction is the incorporation of more granular sub-national or district-level data, which would allow the identification of finer geographic hotspots and more localized drivers of TB burden. Another is the integration of additional health system, behavioral, and environmental variables, including treatment adherence, diagnostic capacity, migration, and urban crowding. Methodologically, future studies could compare ensemble approaches with deep learning, Bayesian hierarchical models, or hybrid explainable AI frameworks to assess whether additional performance gains can be achieved without sacrificing interpretability [8,11]. There is also strong value in expanding GIS-based analysis and spatial epidemiologic modeling to better understand regional clustering and transmission contexts.

In conclusion, this study provides evidence that the global burden of TB and TB-HIV coinfection is driven by overlapping but distinct sets of determinants. TB alone is strongly associated with structural disadvantage and weak health conditions, whereas TB-HIV burden is more directly shaped by HIV epidemiology and the effectiveness of HIV-related care systems [1,4-7]. Machine learning models proved highly effective for capturing these patterns and identifying influential predictors, demonstrating their utility for global health surveillance and decision support [8-10]. Overall, the findings support a comprehensive public health approach in which TB control is linked to broader development goals, and TB-HIV prevention is embedded within integrated HIV care and population health strategies. Such an approach is essential for reducing morbidity and mortality and for making meaningful progress toward global TB elimination targets [1].

## VII. Appendix A: Literature Cited

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## VIII. Appendix B: Code Lines

### Targets

1. `e_inc_100k`
2. `e_mort_100k`
3. `e_inc_tbhiv_100k`
4. `e_mort_tbhiv_100k`

### Required files

Place these CSV files in the same directory as this notebook:

- `WHO-1-TB_burden_countries.csv`
- `WHO-1-MDR_RR_TB_burden_estimates.csv`
- `World-Bank-Development-1.csv`
- `World-Bank-Development-2.csv`

Figure 9: Targets and Required CSV Files

### 1) Load and merge data

```
df_burden = pd.read_csv("/Users/collinlindsay/Desktop/Desktop - Collin's MacBook Pro/CAPSTONE/WHO-1-TB_burden_countries.csv")
df_multidrug = pd.read_csv("/Users/collinlindsay/Desktop/Desktop - Collin's MacBook Pro/CAPSTONE/WHO-1-MDR_RR_TB_burden_estimates.csv")
df_wb_1 = pd.read_csv("/Users/collinlindsay/Desktop/Desktop - Collin's MacBook Pro/CAPSTONE/World-Bank-Development-1.csv")
df_wb_2 = pd.read_csv("/Users/collinlindsay/Desktop/Desktop - Collin's MacBook Pro/CAPSTONE/World-Bank-Development-2.csv")

merged_wb = pd.merge(df_wb_1, df_wb_2, on=["Country", "Country_Code", "Year"], how="inner")
merged_who = pd.merge(df_burden, df_multidrug, on=["Country", "Country_Code", "WHO_Region", "Year"], how="inner")
df = pd.merge(merged_wb, merged_who, on=["Country", "Country_Code", "Year"], how="inner")

print(df.shape)
df.head()
```

✓ 0.0s Python

(1385, 104)

|   | Country     | Country_Code | Year | Access to electricity (% of population) | Adults (ages 15+) and children (ages 0-14) newly infected with HIV | Adults (ages 15-49) newly infected with HIV | Agricultural land (% of land area) | Antiretroviral therapy coverage (% of people living with HIV) | Antiretroviral therapy coverage for PMTCT (% of pregnant women living with HIV) | Arable land (% of land area) | Children (0-14) living with HIV | Children (ages 0-14) newly infected with HIV | Forest area (% of land area) | gross fixed capital formation (constant LCU prices of 2014) |
|---|-------------|--------------|------|-----------------------------------------|--------------------------------------------------------------------|---------------------------------------------|------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------|---------------------------------|----------------------------------------------|------------------------------|-------------------------------------------------------------|
| 0 | Afghanistan | AFG          | 2015 | 71.5                                    | 1100.0                                                             | 1000.0                                      | 58.123668                          | 4.0                                                           | 3.0                                                                             | 11.905309                    | 550.0                           | 200.0                                        | 1.852782                     | 1.45                                                        |
| 1 | Afghanistan | AFG          | 2016 | 97.7                                    | 1100.0                                                             | 1000.0                                      | 58.123668                          | 5.0                                                           | 4.0                                                                             | 11.850114                    | 560.0                           | 200.0                                        | 1.852782                     | 2.260                                                       |
| 2 | Afghanistan | AFG          | 2017 | 97.7                                    | 1000.0                                                             | 1000.0                                      | 58.123668                          | 7.0                                                           | 9.0                                                                             | 11.804118                    | 570.0                           | 100.0                                        | 1.852782                     | 2.647                                                       |
| 3 | Afghanistan | AFG          | 2018 | 93.4                                    | 1100.0                                                             | 1000.0                                      | 58.276988                          | 8.0                                                           | 9.0                                                                             | 11.949772                    | 570.0                           | 100.0                                        | 1.852782                     | 1.189                                                       |
| 4 | Afghanistan | AFG          | 2019 | 97.7                                    | 1200.0                                                             | 1000.0                                      | 58.276988                          | 9.0                                                           | 12.0                                                                            | 11.940573                    | 570.0                           | 100.0                                        | 1.852782                     | 3.911                                                       |

Figure 10: Loading and Merging of Initial Datasets

Missing values (top 25):

|                                                                                                                                             |      |
|---------------------------------------------------------------------------------------------------------------------------------------------|------|
| Community health workers (per 1,000 people)                                                                                                 | 1375 |
| Women making their own informed decisions regarding sexual relations, contraceptive use and reproductive health care (% of women age 15–49) | 1333 |
| Literacy rate, adult total (% of people ages 15 and above)                                                                                  | 1079 |
| PM2.5 air pollution, population exposed to levels exceeding WHO guideline value (% of total)                                                | 884  |
| Births attended by skilled health staff (% of total)                                                                                        | 880  |
| Prevalence of current tobacco use (% of adults)                                                                                             | 814  |
| Antiretroviral therapy coverage for PMTCT (% of pregnant women living with HIV)                                                             | 803  |
| Children (ages 0–14) newly infected with HIV                                                                                                | 792  |
| Children (0–14) living with HIV                                                                                                             | 760  |
| Incidence of malaria (per 1,000 population at risk)                                                                                         | 659  |
| Hospital beds (per 1,000 people)                                                                                                            | 619  |
| Physicians (per 1,000 people)                                                                                                               | 555  |
| Prevalence of severe food insecurity in the population (%)                                                                                  | 500  |
| Incidence of HIV, ages 15–49 (per 1,000 uninfected population ages 15–49)                                                                   | 467  |
| Incidence of HIV, ages 15–24 (per 1,000 uninfected population ages 15–24)                                                                   | 465  |
| Adults (ages 15+) and children (ages 0–14) newly infected with HIV                                                                          | 462  |
| Adults (ages 15–49) newly infected with HIV                                                                                                 | 456  |
| Incidence of HIV, all (per 1,000 uninfected population)                                                                                     | 454  |
| Prevalence of HIV, male (% ages 15–24)                                                                                                      | 399  |
| Prevalence of HIV, female (% ages 15–24)                                                                                                    | 399  |
| Antiretroviral therapy coverage (% of people living with HIV)                                                                               | 395  |
| Prevalence of HIV, total (% of population ages 15–49)                                                                                       | 353  |
| PM2.5 air pollution, mean annual exposure (micrograms per cubic meter)                                                                      | 353  |
| Government expenditure on education, total (% of GDP)                                                                                       | 282  |
| External health expenditure (% of current health expenditure)                                                                               | 195  |
| dtype: int64                                                                                                                                |      |

**Figure 11: Missing Values to Identify Skews**

```
# Numeric correlation with targets
numeric_df = df.select_dtypes(include=[np.number]).copy()

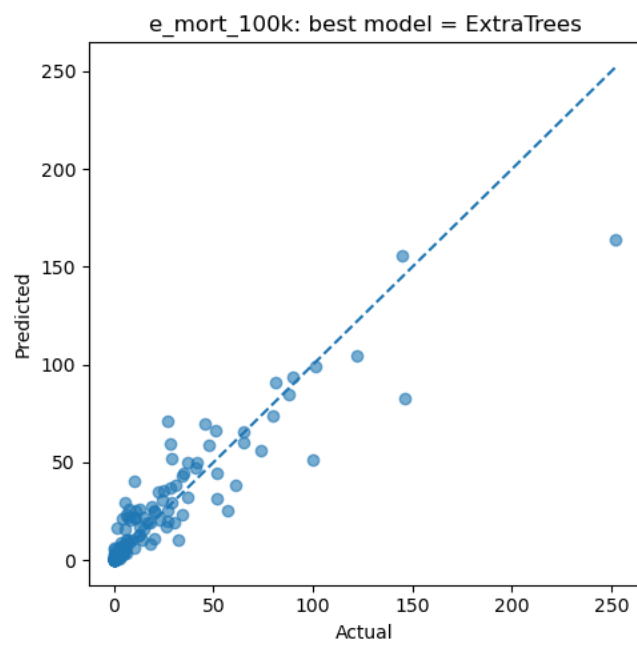
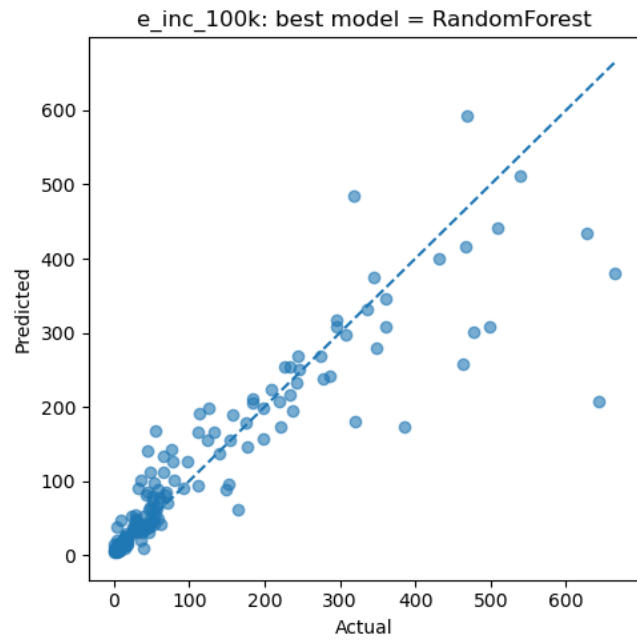
for t in targets:
    corr = numeric_df.corr(numeric_only=True)[t].drop(labels=[t]).sort_values(key=lambda s: s.abs(), ascending=False).head(15)
    print(f"\nTop correlations with {t}:")
    display(corr.to_frame("correlation"))
```

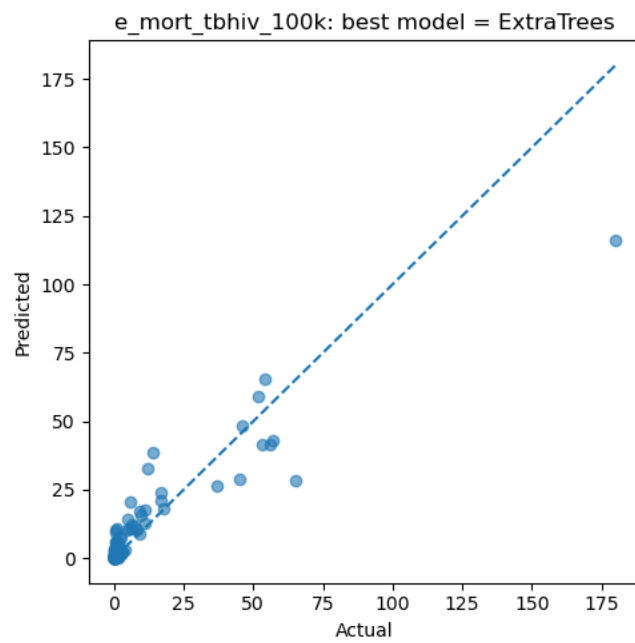
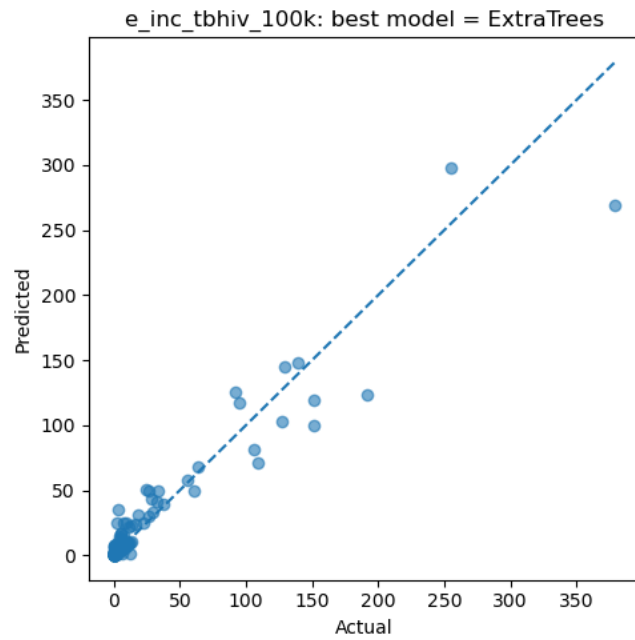
✓ 0.0s Python

Top correlations with e\_inc\_100k:

|                                                                    | correlation |
|--------------------------------------------------------------------|-------------|
| c_newinc_100k                                                      | 0.944536    |
| e_mort_100k                                                        | 0.840359    |
| e_mort_exc_tbhiv_100k                                              | 0.834597    |
| e_inc_tbhiv_100k                                                   | 0.703891    |
| Life expectancy at birth, total (years)                            | -0.629504   |
| e_mort_tbhiv_100k                                                  | 0.628592    |
| Births attended by skilled health staff (% of total)               | -0.518316   |
| Prevalence of severe food insecurity in the population (%)         | 0.506890    |
| Physicians (per 1,000 people)                                      | -0.502825   |
| Adults (ages 15–49) newly infected with HIV                        | 0.458057    |
| Access to electricity (% of population)                            | -0.456605   |
| Adults (ages 15+) and children (ages 0–14) newly infected with HIV | 0.442973    |
| External health expenditure (% of current health expenditure)      | 0.431134    |
| e_inc_tbhiv_num                                                    | 0.429735    |
| Children (0–14) living with HIV                                    | 0.418853    |

**Figure 12: Code to Identify Top Correlations with Target**





```

from sklearn.inspection import permutation_importance

for target in ["e_inc_100k", "e_mort_100k", "e_inc_tbhiv_100k", "e_mort_tbhiv_100k"]:
    target_results = results_df[results_df["Target"] == target].sort_values("Test_RMSE")
    best_name = target_results.iloc[0]["Model"]
    best_model = all_fitted[target][best_name]
    X_train, y_train, X_test, y_test = all_splits[target]

    try:
        imp = permutation_importance(best_model, X_test, y_test, n_repeats=10, random_state=42, n_jobs=-1)
        importances = pd.Series(imp.importances_mean, index=X_test.columns).sort_values(ascending=False).head(15)

        plt.figure(figsize=(8, 5))
        importances.sort_values().plot(kind="barh")
        plt.title(f"Permutation importance: {target} ({best_name})")
        plt.xlabel("Mean importance")
        plt.tight_layout()
        plt.show()

        display(importances.to_frame("importance"))
    except Exception as e:
        print(f"Could not compute permutation importance for {target} / {best_name}: {e}")

```

✓ 22.8s

Python

**Figure 13: Permutation Importance Code Lines**

## IX. Appendix C: GitHub Reference and Raw Data

<https://github.com/clindsay0258-wq/Meharry-Predicting-Global-and-Regional-Tuberculosis-Trend>